

```
CREATE TYPE Account_t_IS1 AS OBJECT (  
    accNo INTEGER,  
    accName VARCHAR2(50),  
    balance FLOAT  
)  
NOT FINAL NOT INSTANTIABLE;  
/  

```

```
CREATE TYPE Savings_t_IS1 UNDER Account_t_IS1 (  
    interest_rate FLOAT  
);  
/  

```

```
CREATE TYPE Current_t_IS1 UNDER Account_t_IS1 (  
    No_cheques_issued INTEGER  
);  
/  

```

```
CREATE TYPE Account_Ref_t_IS1 AS OBJECT (  
    account REF Account_t_IS1  
);  
/  

```

```
CREATE TYPE Account_Ref_tbl AS TABLE OF Account_Ref_t_IS1;  
/  

```

```
CREATE TYPE Customer_t_IS1 AS OBJECT (  
    nic CHAR(10),  
    name VARCHAR2(50),  
    accounts Account_Ref_tbl  
);  

```

```
);  
/
```

```
CREATE TABLE Accounts_IS1 OF Account_t_IS1 (PRIMARY KEY(accNo));
```

```
CREATE TABLE Customers_IS1 OF Customer_t_IS1 (PRIMARY KEY (nic)) NESTED TABLE accounts  
STORE AS st_acc_tbl;
```

```
INSERT INTO Accounts_IS1 VALUES (Savings_t_IS1(10, 'Ran Surakum', 1000000, 4.5));
```

```
INSERT INTO Accounts_IS1 VALUES (Current_t_IS1(4, 'Normal Checking', 23490.0, 156));
```

```
INSERT INTO Accounts_IS1 VALUES (Savings_t_IS1(2, 'Ran Surakum', 12000, 4.5));
```

```
INSERT INTO Accounts_IS1 VALUES (Current_t_IS1(12, 'Business Account', 234521.33, 1347));
```

```
INSERT INTO Customers_IS1 VALUES ('1234567890', 'Mr. Suraweerasaratne',
```

```
Account_Ref_tbl (
```

```
Account_Ref_t_IS1((select REF(a1) from Accounts_IS1 a1 where a1.accNo = 10)),
```

```
Account_Ref_t_IS1((select REF(a2) from Accounts_IS1 a2 where a2.accNo = 4))
```

```
)
```

```
);
```

```
INSERT INTO Customers_IS1 VALUES ('1232267890', 'Mr. Peiris',
```

```
Account_Ref_tbl(
```

```
Account_Ref_t_IS1((select REF(a) from Accounts_IS1 a where a.accNo = 2))
```

```
)
```

```
);
```

```
INSERT INTO Customers_IS1 VALUES ('1232288890', 'Mr. Wijetunga',
```

```
Account_Ref_tbl(
```

```
Account_Ref_t_IS1((select REF(a) from Accounts_IS1 a where a.accNo = 12))
```

```
)
```

);

Question 01:

Get Customer name and nic of customers with atleast one account.

```
SELECT nic, name  
FROM Customers_IS1 c  
WHERE EXISTS (SELECT 1 FROM TABLE(c.accounts));
```

Question 02:

Create function to return total minimum balance of an account.

To create a function that returns the total minimum balance of an account, you can iterate over the accounts belonging to a customer and calculate the minimum balance for each account. Here's the function:

```
CREATE OR REPLACE FUNCTION get_total_min_balance(nic_in CHAR) RETURN  
FLOAT IS  
    total_min_balance FLOAT := 0;  
BEGIN  
    FOR acc_ref IN (  
        SELECT a.account  
        FROM Customers_IS1 c, TABLE(c.accounts) a  
        WHERE c.nic = nic_in  
    ) LOOP
```

```

total_min_balance := total_min_balance + (
    SELECT MIN(CASE
        WHEN a.accName = 'Savings' THEN CAST(((a.balance *
a.interest_rate) / 100) AS FLOAT)
        ELSE a.balance
    END)
    FROM Accounts_IS1 a
    WHERE Deref(acc_ref.account).accNo = a.accNo
);
END LOOP;

RETURN total_min_balance;
END;
/

```

This function takes a NIC (customer ID) as input and returns the total minimum balance for all accounts associated with that customer. It iterates over the accounts using a cursor and calculates the minimum balance for each account, considering whether it's a savings account or not. Finally, it returns the total minimum balance.

Question 03:

Print customer name and minimum balance in descending order.

```
SELECT c.name AS customer_name,  
       MIN(CASE  
           WHEN Deref(a.account).accName = 'Savings' THEN  
             CAST(((Deref(a.account).balance * Deref(a.account).interest_rate) / 100) AS  
               FLOAT)  
           ELSE Deref(a.account).balance  
         END) AS minimum_balance  
FROM Customers_IS1 c  
JOIN TABLE(c.accounts) a ON 1=1  
GROUP BY c.name  
ORDER BY minimum_balance DESC;
```